# Variant-Calling Analysis Report

## 1.0 Introduction

This report documents a short-read variant-calling and functional-annotation workflow applied to three whole-genome resequencing samples drawn from population Ara-3 of Richard Lenski's Escherichia coli Long-Term Evolution Experiment (LTEE), corresponding to evolutionary generations 5,000 (sample Ara3_5000, SRA run SRR2589044), 15,000 (Ara3_15000, SRR2584863), and 50,000 (Ara3_50000, SRR2584866). Paired-end 150 bp Illumina reads (1.112.77 million read pairs per sample) were quality-trimmed, aligned to the ancestral REL606 reference genome, and used to call haploid variants.

Mapping rates exceeded 99.9% across all three samples, with mean genome-wide depth ranging from X50.8 to X119.2. Quality-filtered variant counts rose sharply with generation number (9 PASS variants at generation 5,000, 26 at generation 15,000, and 794 at generation 50,000) a roughly 30-fold increase between the second and third timepoints consistent with the previously reported evolution of a hypermutator phenotype in this lineage. The transition/transversion (Ts/Tv) ratio follows the same trajectory (2.00 → 0.33 → 10.86), shifting from a near-random mutational signature at generation 15,000 to a strongly transition-biased signature at generation 50,000.

Functional annotation with SnpEff against a custom REL606 gene model classified the majority of variants as MODIFIER-impact at every timepoint, with missense, frameshift, and stop-gained variants accumulating specifically by generation 50,000 (369, 62, and 9 occurrences, respectively). Two loci of direct biological interest recur in the annotated output: the citrate-utilization (cit) operon, which carries missense and regulatory variants exclusively at generation 50,000, and the DNA mismatch-repair gene set (mutS, mutL, mutY, dnaQ, recA/B/C, dam), which includes a premature stop codon in mutS (p.Gln606*) at generation 50,000. Both findings are congruent with the independently documented evolutionary history of population Ara-3.

## 2.0 Methods

### 2.1 Data Acquisition

Three paired-end FASTQ files were retrieved from SRA.

### 2.2 Quality Control and Trimming

Raw and post-trimming quality control were performed with FastQC followed by MultiQC aggregation. Adapter and quality trimming used Trimmomatic in paired-end. Paired-read retention ("both surviving") was 79.26% (SRR2589044), 81.37% (SRR2584863), and 73.01% (SRR2584866).

### 2.3 Reference Genome Preparation

The ancestral REL606 genome (GCA_000017985.1_ASM1798v1, a single 4,629,812 bp chromosome) was obtained from NCBI, indexed with bwa index, and indexed with samtools faidx.

### 2.4 Alignment

Trimmed reads were aligned with BWA-MEM against REL606 and piped directly into samtools sort to generate coordinate-sorted BAM files. Each BAM was verified with samtools quickcheck, indexed, and summarized with samtools flagstat, samtools stats, and samtools depth -aa (genomewide, zero-filled).

**Table 1. Alignment quality metrics per sample (samtools flagstat / stats / depth)**

| Sample | Generation | Total Reads | Mapping Rate | Properly Paired | Mean Depth | Breadth ≥10× |
|---|---|---|---|---|---|---|
| Ara3_5000 | 5,000 | 1,774,505 | 99.94% | 95.34% | 50.77× | 99.83% |
| Ara3_15000 | 15,000 | 2,541,520 | 99.96% | 97.64% | 72.67× | 99.84% |
| Ara3_50000 | 50,000 | 4,056,102 | 99.98% | 98.73% | 119.16× | 93.07% |

### 2.5 Variant Calling

Genotype likelihoods were generated with bcftools mpileup and haploid variants called with bcftools call, reflecting the clonal, asexual nature of the *E. coli* populations under study:

```
bcftools mpileup --threads 4 -Ou -f <ref.fasta> \
    -a FORMAT/AD,FORMAT/DP <bam> | bcftools view -Ob -o <run>.raw.bcf

bcftools call --threads 4 --ploidy 1 -m -v -Oz \
    -o <run>.raw.vcf.gz <run>.raw.bcf

bcftools filter --threads 4 \
    -e 'QUAL<20 || FORMAT/DP<10' -s LowQual -Oz \
    -o <run>.filtered.vcf.gz <run>.raw.vcf.gz
```

PASS-only extraction (bcftools view -f PASS) and normalization (bcftools norm -f <ref> -m -any) prepare the input consumed by the annotation stage.

### 2.6 Functional Annotation

Normalized PASS variants were annotated with SnpEff against a custom-built REL606 gene model (4,209 of 4,209 predicted proteins passed the SnpEff build-time consistency check, 0 errors).

## 3. Variant Analysis Results & Annotation Summary

### 3.1 Variant Counts by Type

**Table 2. Raw versus quality-filtered (PASS) variant counts, composition, and Ts/Tv ratio per sample**

| Sample | Generation | Raw Variants | PASS Variants | PASS SNPs | PASS Indels | Ts/Tv |
|---|---|---|---|---|---|---|
| Ara3_5000 | 5,000 | 13 | 9 | 6 | 3 | 2.00 |
| Ara3_15000 | 15,000 | 30 | 26 | 20 | 6 | 0.33 |
| Ara3_50000 | 50,000 | 817 | 794 | 688 | 106 | 10.86 |

### 3.2 Mutational Spectrum

The Ts/Tv ratio moves from 2.00 (generation 5,000) to 0.33 (generation 15,000, transversiondominated) to 10.86 (generation 50,000, strongly transition-dominated). At generation 50,000, the collapsed single-nucleotide substitution spectrum is dominated by C>T (370 events) and A>G (260 events) changes whereas generation 15,000 is dominated by A>C (8) and C>A (6) transversions on a much smaller total (20 SNPs).

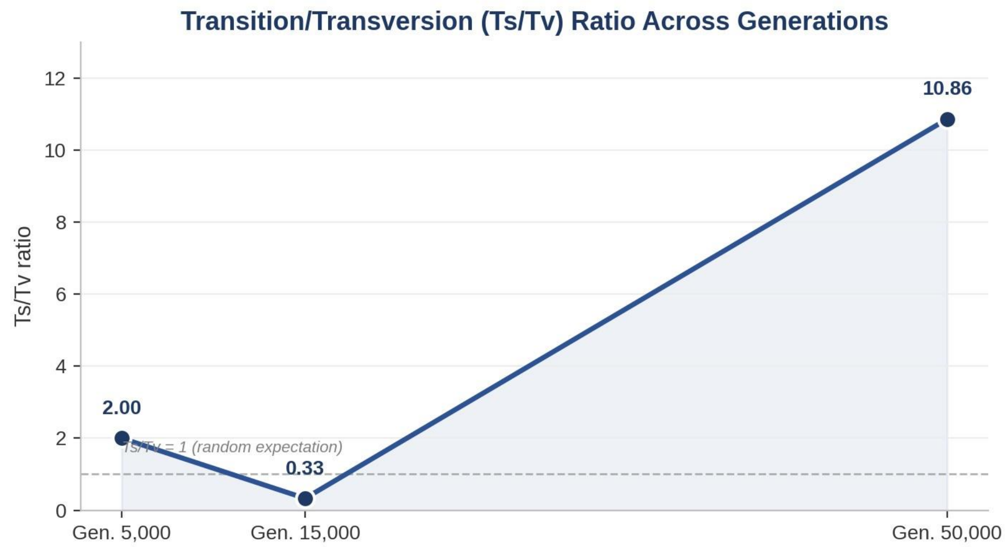


Figure 1. Transition/transversion (Ts/Tv) ratio across the three sampled generations.

### 3.3 Functional Impact Annotations

**Table 3. SnpEff impact-category counts by generation**

| Generation | High | Moderate | Low | Modifier |
|---|---|---|---|---|
| 5,000 | 2 | 6 | 0 | 75 |
| 15,000 | 3 | 17 | 0 | 232 |
| 50,000 | 72 | 374 | 213 | 7,367 |

**Table 4. Selected SnpEff consequence-type counts by generation (annotation records per affected transcript/feature)**

| Consequence | Gen. 5,000 | Gen. 15,000 | Gen. 50,000 |
|---|---|---|---|
| Missense variant | 6 | 16 | 369 |
| Synonymous variant | 0 | 0 | 212 |
| Frameshift variant | 2 | 3 | 62 |
| Stop-gained (nonsense) | 0 | 0 | 9 |
| Intergenic region | 1 | 6 | 122 |
| Upstream / downstream gene variant | 74 | 224 | 7,228 |

### 3.4 High-Impact Target Genes and Recurrent Hot Spots

High-impact (HIGH-category: frameshift, stop-gained) variant counts were 2, 3, and 72 for generations 5,000, 15,000, and 50,000 respectively. The genes with the highest per-generation annotation-record counts at generation 50,000 were yhhH, rhsB, nikR, nikE, and nikD (13 records each), followed by yhhI (10) and the ygcS/ycdW/nikB/nikA/deoA group (8 each). The recurring nik operon (nickel-transport) cluster likely reflects several closely spaced variants annotated against overlapping upstream/downstream windows of adjacent nik genes.

*3.4.1 Citrate-Utilization Operon*

Variants in the citT/citF/citA/citG/citE operon (and the linked dctA/rnk region) appear exclusively in the generation-50,000 sample. These include a missense change in citT (p.Thr341Ile, the citrate/succinate antiporter), a missense change in citF (p.Glu51Lys, citrate lyase subunit), and a frameshift in citA (p.Val493fs). No cit-operon variants are present at generations 5,000 or 15,000.

*3.4.2 DNA Mismatch-Repair and Replication-Fidelity Genes*

A single missense variant in dam (p.Pro141Thr) is present at generation 5,000. By generation 50,000, this gene set carries additional variants, most notably a premature stop codon in mutS (c.1816C>T, p.Gln606*) and a missense change in mutS itself (p.Arg500His), alongside missense variants in mutL (p.Ser567Gly) and recC (p.Val1081Ala).

**Table 5. Selected high-impact and biologically flagged variants in mismatch-repair and citrate-operon genes**

| Sample | Gene | Consequence | Impact | Protein Change |
|---|---|---|---|---|
| Ara3_50000 | mutS | Stop-gained | HIGH | p.Gln606* |
| Ara3_50000 | mutS | Missense | MODERATE | p.Arg500His |
| Ara3_50000 | mutL | Missense | MODERATE | p.Ser567Gly |
| Ara3_5000 | dam | Missense | MODERATE | p.Pro141Thr |
| Ara3_50000 | citT | Missense | MODERATE | p.Thr341Ile |
| Ara3_50000 | citF | Missense | MODERATE | p.Glu51Lys |
| Ara3_50000 | citA | Frameshift | HIGH | p.Val493fs |
| Ara3_50000 | mutY | Frameshift | HIGH | p.Gly318fs |
| Ara3_50000 | lacZ | Stop-gained | HIGH | p.Trp433* |
| Ara3_50000 | ddlB | Stop-gained | HIGH | p.Trp182* |

## 4. Discussion & Biological Relevance

Population Ara-3 is one of twelve replicate *E. coli* B populations founded from a common REL606 ancestor at the start of the LTEE in 1988, and is the population in which two of the most widely studied evolutionary events in the experiment's history occurred. The appearance of a hypermutator (mutator) phenotype and, later, the evolution of aerobic citrate utilization (Cit+). The variant data recovered here are directly consistent with both published events, even though the pipeline itself contains no explicit knowledge of LTEE history. The signal emerges purely from the raw counts, the Ts/Tv trajectory, and the annotated gene identities.

### 4.1 Mutator Phenotype

The nearly 30-fold increase in PASS variant count between generations 15,000 and 50,000 (26 → 794), combined with the swing in Ts/Tv ratio from a transversion-biased 0.33 to a strongly transition-biased 10.86, is the expected molecular fingerprint of a loss-of-function event in the mismatch-repair (MMR) pathway. The premature stop codon identified in mutS (p.Gln606*) at generation 50,000 is a plausible candidate lesion consistent with this signature: MutS initiates mismatch recognition in the MutHLS repair pathway, and its inactivation is a well-established route to elevated point-mutation rates in *E. coli*, producing exactly the kind of transition-enriched, genome-wide accumulation of missense, synonymous, and stop-gained substitutions observed here. Additional variants in mutL, dnaQ (the proofreading subunit of DNA polymerase III), and recA/recB/recC by generation 50,000 further support a broader erosion of replication-fidelity and repair function in this lineage by the later timepoint. Establishing causal loss-of-function for any single variant would require complementation or structural analysis beyond the scope of this variant-calling pipeline.

### 4.2 Citrate Utilization (Cit+)

The appearance of missense and frameshift variants in the citT/citF/citA/citG/citE operon strictly at generation 50,000, with no such variants detectable at 5,000 or 15,000 generations, is consistent with the historically documented emergence of the Cit+ trait in population Ara-3 at approximately generation 31,500. CitT (the citrate/succinate antiporter) and CitF (a citrate lyase subunit) are both components of the anaerobic citrate fermentation system that, through a genomic rearrangement placing citT under aerobic-active regulatory control, was shown to enable this population's now well-known ecological innovation. The point substitutions recovered here should be interpreted as co-occurring background mutations accumulated in and around the cit operon by generation 50,000, not necessarily as the causal Cit+ rearrangement itself, since a short-read bcftools-based SNP/indel caller without dedicated structural-variant detection is not well positioned to resolve the underlying tandem duplication reported in the primary literature.

### 4.3 Overall Interpretation

Considered together, the rising raw and PASS variant counts, the inverted Ts/Tv trajectory, the concentration of HIGH- and MODERATE-impact annotations at the latest timepoint, and the identity of the specific genes affected (DNA-repair machinery and the citrate operon) converge on a coherent and internally consistent evolutionary narrative for this lineage, even though the pipeline draws on no external LTEE-specific reference data beyond the REL606 genome and its GFF-derived gene model. This consistency between an annotation-agnostic variant-calling workflow and the independently documented biological history of population Ara-3 supports the pipeline's reliability, as configured (haploid calling, QUAL≥20/DP≥10 PASS filtering, SnpEff functional annotation), for detecting biologically meaningful signal in microbial experimentalevolution resequencing data.

### 4.4 Limitations

All three samples represent shallow, single-replicate snapshots at their respective generations rather than population-level allele-frequency time series.

No formal statistical association between specific variants and phenotype was conducted.